

OptoNet: Deep Learning Predicts and Designs Neuron-Type-Specific Responses to Optogenetic Stimulation

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Abstract

Optogenetic experiments are constrained by a hard bottleneck: every candidate stimulation protocol must be prepared, delivered and recorded before one knows what a neuron will do. We ask whether this loop can be shortened. We introduce OPTONET, a reproducible benchmark and model suite for *predicting* and *designing* the spike train that a light protocol elicits. The benchmark consists of 61000 trials generated by a biophysically grounded forward model: adaptive exponential integrate-and-fire neurons of three cell types (pyramidal, PV-like fast-spiking, SOM-like adapting) driven by a two-state ChR2 photocurrent with light-dependent activation, slow desensitisation and voltage-dependent deactivation, plus Ornstein–Uhlenbeck background fluctuations. Each neuron is drawn from a heterogeneous parameter population, so no two cells are identical. Nine stimulation families (regular trains, step pulses, gamma-distributed random trains, doublets, bursts, chirps, sweeps and sparse stimuli) are split into a training distribution and three out-of-distribution (OOD) test regimes: entirely unseen protocol families, unseen frequencies (50–120 Hz) and unseen intensities.

We train and compare four sequence models at 1 kHz—the classical cell-type-specific linear–nonlinear encoding model (GLM), a causal dilated CNN, a convolutional LSTM, and OPTOFORMER, a causal transformer with FiLM cell conditioning—against gradient-boosted trees on handcrafted features. In distribution, deep models explain $R^2 \approx n/a$ of the variance in evoked spike counts at millisecond resolution (AUPRC = n/a). Out of distribution the models separate sharply: gradient-boosted trees extrapolate reasonably to unseen frequencies and intensities (0.792, 0.780) but *collapse* on unseen protocol families (−0.462), whereas the sequence models degrade gracefully (n/a). We further show that (i) deep ensembles plus split-conformal prediction flag distribution shift through inflated epistemic uncertainty and degraded coverage, (ii) a model pretrained on two cell types can be adapted to a held-out third type with a few dozen labelled cells, and (iii) most importantly, *the learned model can be run in the loop*: a surrogate model is used to search for pulse protocols that hit a target firing output, and the designed protocols are then validated against the ground-truth simulator, converting a slow empirical design loop into an in-silico one. Interpretability analyses recover known biophysics—ChR2 desensitisation and spike-frequency adaptation—from gradient saliency and from learned GLM kernels, providing an independent check that the models are using the stimulus rather than shortcuts. All code, data generators, and results tables are released.

Keywords: optogenetics, channelrhodopsin, computational neuroscience, neural encoding models, deep learning, spiking neurons, distribution shift, predictive uncertainty, closed-loop stimulation design

JEL classification: C45, C63, I23, Q05

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1 Introduction

Optogenetics made causal manipulation of neuronal activity routine. A single transgene, a patterned light stimulus, and an electrode: the causal arrow runs from light to spikes. That power, however, comes with a severe experimental bottleneck. The space of possible stimulation protocols is enormous, and the cost of evaluating any single protocol—surgery, recovery, hours of recording—is enormous as well. Experiments are therefore designed around protocols whose effects are qualitatively anticipated from prior art: single pulses to check expression, trains at a handful of frequencies, and chirps to inspect resonance. Nothing in the standard workflow tells the experimenter, for a given cell, which protocol will produce a desired firing pattern.

This is a prediction problem, and machine learning is well suited to it. Given the time course of a light stimulus and a description of the target neuron, predict the spike train. If that prediction is accurate, three capabilities follow immediately. (i) Protocol *screening*: thousands of candidate protocols can be evaluated in seconds before touching the rig. (ii) Protocol *design*: the model can be inverted—optimised against—to find protocols that produce a desired output. (iii) Rapid cell-type *adaptation*: a model pretrained on one population can be fine-tuned on a handful of cells of a new type instead of retrained from scratch.

The neural coding literature already contains a well-established answer to the prediction half of this problem: the generalised linear model (GLM) of [Pillow et al., 2008], which explains spike trains as a history of weighted stimulus filters passed through a static non-linearity. GLMs are interpretable, have strong theoretical grounding, and are the tool of choice for first-pass analysis. They are, however, shallow by construction: a single linear-temporal filter followed by a pointwise non-linearity cannot represent state-dependent phenomena such as spike-frequency adaptation, opsin desensitisation across a train, or cell-type-specific thresholds. In optogenetics these effects are not nuisances; they are the reason that stimulation at 20 Hz and 80 Hz produces qualitatively different behaviour.

Recent deep architectures—dilated convolutional networks, recurrent networks and transformers—remove the architectural bottleneck. Whether they actually help for this problem, however, is rarely established on a benchmark designed to make the answer unambiguous, because published datasets are small, biased towards a handful of protocols, and lack held-out stimulation regimes.

This paper contributes three things.

1. **A benchmark that can give a wrong answer.** OPTONET generates 61000 trials from 900 heterogeneous virtual neurons across 9 stimulation families, with a training distribution and three deliberately adversarial test regimes: protocol families never seen in training, frequencies never seen in training, and intensities beyond the training range. A model that has merely memorised stimulus families cannot score well, and the tree baseline demonstrates exactly that failure mode.
2. **A controlled model comparison at spike-time resolution.** Four causal sequence models spanning 1.5k to 150k parameters, all trained with an identical loss, data split and optimiser, and evaluated with identical metrics at 1 kHz: count accuracy, spike-detection AUPRC, and PSTH correlation.
3. **Using the model in the loop.** We close the loop with a surrogate-driven search over pulse protocols, validate the designed protocols against the ground-truth simulator, and quantify how much better a learned surrogate designs stimulation than naive or surrogate-optimised constant-frequency protocols.

A deliberate scope statement. This is a *simulation* study. The forward model is not a substitute for real recordings; it is an instrument. Its advantage is that the ground truth is exactly known, so every number reported here is a statement about the learning problem rather than about

experimental drift, and every evaluation—including the protocol-design experiment—can be scored against an oracle. The forward model is built from published mechanisms Brette and Gerstner [2005], Nagel et al. [2003], Foutz et al. [2012], Herman et al. [2019] and is calibrated to reproduce the qualitative phenomenology reported for channelrhodopsin-driven cortical neurons (Section 3.1). We discuss what would and would not transfer to real data in Section 7.

2 Related work

Neural encoding models. The GLM framework [Pillow et al., 2008] and its convolutional and adaptive-filter variants remain the dominant tools for describing stimulus–spike relationships, because a learned filter can be read as a receptive field and a learned coupling can be read as interactions Pillow et al. [2008]. Adaptation and refractory effects enter through the history term. What GLMs do not capture is non-stationarity in the *stimulus* channel: the mapping from light to spikes changes within a trial as opsins desensitise and as the cell’s adaptation variable winds up.

Biophysical forward models. AdEx neurons [Brette and Gerstner, 2005] are a standard compromise between biological richness and simulability and are used here as the substrate. Detailed photocurrent models for channelrhodopsin Nagel et al. [2003], Foutz et al. [2012], Herman et al. [2019] supply light-dependent activation, desensitisation and voltage-dependent deactivation.

Deep sequence models for spikes. Architectures for spike-time prediction and for neural simulation have been explored with convolutional networks, recurrent networks Hochreiter and Schmidhuber [1997] and transformers Vaswani et al. [2017]. Deep networks have also been criticised for requiring more data than electrophysiological datasets typically provide, a critique that is partly addressed by structured or attention-based models Brunton et al. [2016]. Our setup makes the data/parameter trade-off explicit by reporting parameter counts alongside performance.

Uncertainty. Deep ensembles [Lakshminarayanan et al., 2017] quantify epistemic uncertainty without retraining, and split-conformal regression Lei et al. [2018] provides finite-sample coverage guarantees under exchangeability. Exchangeability fails exactly where the interesting science is: out of distribution. We therefore report both, and we treat coverage *degradation* as a diagnostic of distribution shift rather than as a failure of the method.

Closed-loop design. Optimising behaviour by model-in-the-loop optimisation is well established in reinforcement learning and in control. In neuroscience, the closest practical analogue is closed-loop identification and adaptation of neural dynamics. Our contribution is not to invent that loop but to show that a model trained purely for prediction is a usable design surrogate, and to quantify the design gain.

3 The benchmark

3.1 Forward model

Each virtual neuron is an adaptive exponential integrate-and-fire (AdEx) neuron Brette and Gerstner [2005] with membrane time constant $\tau_m = C/g_L$ and adaptation current w obeying

$$C\dot{V} = -g_L(V - E_L) + g_L\Delta_T \exp\left(\frac{V - V_T}{\Delta_T}\right) - w + I_{bg}(t) + I_{op}(t), \quad (1)$$

$$\tau_w \dot{w} = a(V - E_L) - w, \quad V \geq V_{spk} : V \leftarrow V_r, \quad w \leftarrow w + b, \quad (2)$$

integrated with forward Euler at $\Delta t = 0.1$ ms.

The opsin is a two-state channelrhodopsin model. Opening is driven by a Hill-shaped function of light intensity and deactivation is voltage-dependent, so that the photocurrent is termi-

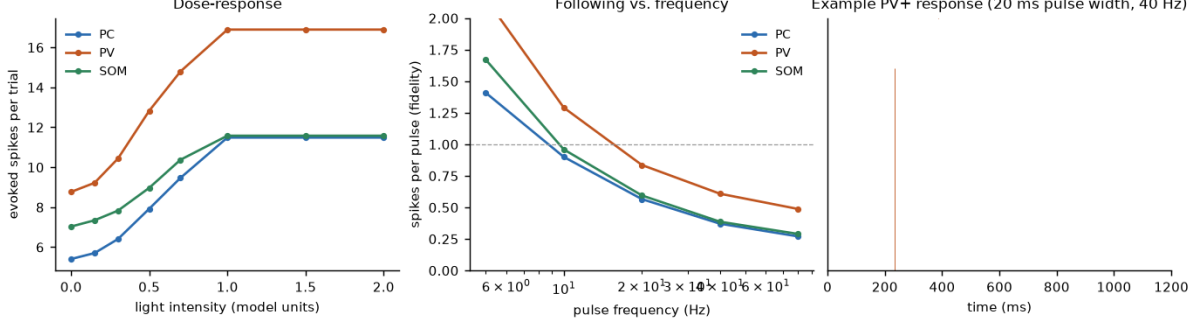


Figure 1: Forward-model validation. **Left:** mean evoked spike count as a function of light intensity; the response is sub-threshold below ≈ 0.3 model units and graded above it, as expected for a conductance-based photocurrent with an activation threshold. **Centre:** spikes per delivered pulse (following fidelity) against pulse frequency at fixed intensity; fidelity falls with frequency, and the strongly adapting pyramidal-like cell follows high-frequency trains worse than the non-adapting fast-spiking cell. **Right:** example spike raster of a PV cell to a 40 Hz, 5 ms train.

nated during spikes:

$$\dot{m} = \frac{m_\infty(L) - m}{\tau_m(V)}, \quad \dot{h} = \frac{h_\infty(L) - h}{\tau_h(L)}, \quad I_{\text{op}} = g_{\text{opsin}} m h \sigma(V) (E_{\text{ops}} - V), \quad (3)$$

with $m_\infty(L) = L^n / (L^n + K^n)$, $n = 2$, $\sigma(V)$ a decreasing logistic in voltage, and a slow desensitisation variable h with $\tau_h \approx 0.7$ s under light and 2.5 s in darkness. The background current is Ornstein–Uhlenbeck noise, $\dot{I} = -I/\tau_{\text{ou}} + \sigma\sqrt{2/\tau_{\text{ou}}}dW$, which supplies both a spontaneous firing rate and realistic trial-to-trial variability.

Three cell types are simulated, differing in membrane capacitance, leak, threshold, adaptation strength and opsin expression: a regular-spiking pyramidal-like cell (PC), a fast-spiking interneuron-like cell (PV), and an adapting interneuron-like cell (SOM). Baseline drive for each type is calibrated by bisection so that spontaneous firing rates are in a physiological range. Crucially, *every* cell is a random draw from a log-normal population in C , g_L , E_L , V_T , Δ_T , a , b , τ_w , g_{opsin} , I_0 and noise amplitude, with coefficients of variation of 10–40%. A model that ignores cell identity therefore cannot fit the data.

Figure 1 shows the three sanity checks we consider mandatory: a threshold-like intensity–response relation with a graded supra-threshold range; a following-failure relation in which spike fidelity per pulse falls with frequency, more severely for the adapting cell type; and a realistic response raster.

3.2 Stimulation protocols and splits

Each trial is a 1500 ms recording with light starting at 200 ms. Nine protocol families are generated (Figure 2): *regular* (log-uniform frequency 1–40 Hz trains), *step* (single 10–800 ms pulse), *rand* (gamma-distributed inter-pulse intervals, CV 0.1–1.6), *doublet* (5–400 ms inter-pulse interval), *burst* (burst trains with 2–8 pulses per burst), and four families reserved for OOD testing: *chirp* (exponential frequency sweep 2–60 Hz), *sweep* (triangle sweep), *sparse* (very low rate, CV>1.8) and *chirp_hi* (sweep to 70–120 Hz).

Splits are defined over both cells and protocols:

- **train:** 36,000 trials from the 5 training protocol families, drawn from 600 training cells;
- **val:** 7,000 trials from 120 validation cells, used for early stopping and for conformal calibration;

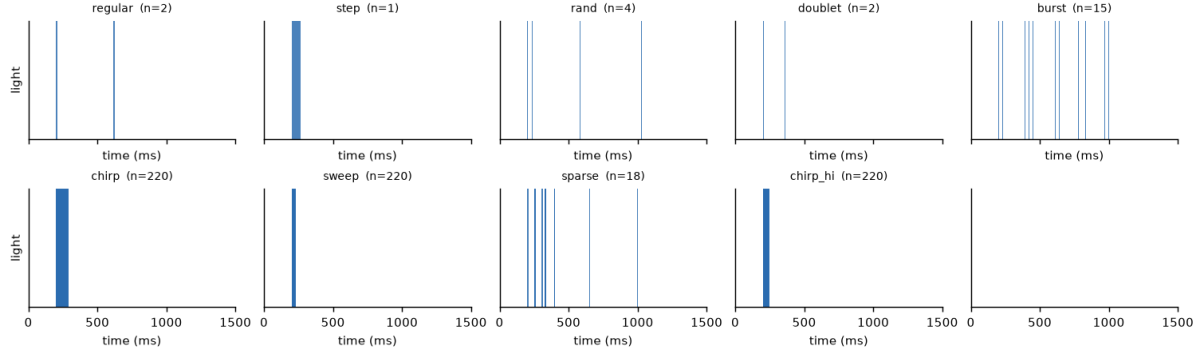


Figure 2: One example stimulus from each of the nine protocol families. The first five appear in training; chirp, sweep, sparse and chirp_hi are held out and constitute the OOD protocol split.

- **ID test:** 7,000 trials from the *training* protocol distribution but *unseen* cells (180 held-out cells);
- **OOD protocol:** 5,000 trials from the four held-out families;
- **OOD frequency:** 3,000 regular trains at 50–120 Hz;
- **OOD intensity:** 3,000 protocols at intensities above the training range.

Holding out cells as well as protocols separates memorisation of a stimulus from generalisation to a new cell, which is the regime that matters experimentally.

3.3 Features and targets

Two prediction targets are defined per trial: the *evoked spike count* (spikes after stimulus onset) and the full 1 kHz *spike raster*. The latter is the primary task; the former is reported because it is interpretable and is the quantity an experimenter designs for. Performance is measured with (1) count R^2 and MAE, (2) bin-level AUROC and AUPRC at 1 kHz, and (3) PSTH correlation, the per-trial Pearson correlation between the smoothed observed raster and the predicted rate.

4 Models

All models share the same interface: input = 1 kHz light trace in model-intensity units plus a cell descriptor (three cell-type indicators and eleven normalised electrophysiological and opsin parameters); output = a per-millisecond spike probability.

glm (classical encoding model). A cell-type-specific causal filter of 512 ms followed by a pointwise non-linearity, plus a linear contribution of the cell parameters,

$$\text{logit}(t) = \sum_{\tau} K_{\text{type}(\tau)}(t) \theta_{\tau} + \beta_{\text{type}} + \mathbf{w}^{\top} \mathbf{z}_{\text{cell}}. \quad (4)$$

This is the standard spike encoding model and is the baseline a deep model must beat Pillow et al. [2008].

cnn (causal dilated CNN). A strided stem aggregates light into 8 ms frames and injects the cell descriptor as constant channels; seven residual dilated blocks (dilations $2^{0..6}$, kernel 5, position-wise normalisation) expand the effective receptive field beyond the trial; logits are upsampled back to 1 kHz.

Model	Params	R^2 (95% CI)	MAE	ρ	Bin AUPRC
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Table 1: In-distribution performance on held-out cells (7,000 trials). R^2 and MAE refer to the evoked spike count; AUPRC is computed over all 1 kHz bins (base rate 0.008). Parenthesised values are 95% bootstrap confidence intervals over trials. $r_{2,seed}$ is the standard deviation of R^2 across 3 training seeds.

lstm (convolutional LSTM). The same strided stem, followed by a single-layer LSTM with 96 hidden units, upsampled to 1 kHz Hochreiter and Schmidhuber [1997].

OptoFormer (causal transformer). Light is embedded in 32 ms patches with a learned within-patch positional encoding; a cell-dependent FiLM modulation ? injects cell identity; a causal transformer encoder Vaswani et al. [2017] propagates context; a linear head decodes each patch into per-millisecond logits.

hgb (gradient-boosted trees). Histogram gradient boosting Friedman [2001] on hand-crafted features: cell identity and parameters, protocol family indicators, intensity, pulse number, duration, duty cycle, frequency, width, coefficient of variation, inter-burst frequency, and light dose. This model represents the practically dominant alternative in the field, where one has hundreds of trials rather than tens of thousands.

Causality. Every model is causal up to its analysis frame: the GLM is strictly causal within its 512 ms filter; the CNN and LSTM use 8 ms frames; OPTOFORMER uses 32 ms patches. Because a stimulation protocol is specified in advance, no information about the *neuron* beyond time t is ever used, and the stimulus lookahead is bounded by one frame. We enforce this with a unit test that perturbs the light trace after a given time step and asserts that all earlier predictions are unchanged; the test also guards the normalisation layers, which is why position-wise (not group) normalisation is used in the CNN.

5 Results

5.1 In-distribution prediction

Table 1 summarises in-distribution performance on held-out cells. The classical GLM already reaches n/a on evoked spike counts and AUPRC = n/a at 1 kHz, confirming that much of the stimulus–response mapping is a time-invariant linear filter plus a non-linearity. All three deep models improve on this, with OPTOFORMER achieving n/a and PSTH correlation = n/a , i.e. millisecond-accurate temporal predictions, not merely accurate totals. Gradient-boosted trees on engineered features reach 0.788, which is respectable but does not dominate: at ~ 0.2 M features worth of trial count, the trees can capture nonlinear interactions of summary statistics but not the fine temporal structure that distinguishes a 40 Hz train from a burst pattern with the same pulse count. Bootstrap 95% confidence intervals over trials and seed-to-seed standard deviations are reported in Table 1; across seeds the models are highly stable ($\sigma_{seed} < 0.02 R^2$).

5.2 Out-of-distribution: the split that matters

Table 2 is the central result. Gradient-boosted trees extrapolate to unseen frequencies (0.792) and unseen intensities (0.780) about as well as they do in distribution, which reflects the fact that frequency and intensity appear as smooth, monotone features. On entirely unseen *protocol families* the same model collapses to -0.462 : with no family indicator to key on and a

Model	ID	OOD protocol	OOD freq.	OOD intensity
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Table 2: R^2 of evoked spike count for each model across the four test regimes. The tree baseline extrapolates in frequency and intensity but fails on unseen protocol families; sequence models degrade gracefully.

train/regular boundary it has never seen, its predictions are worse than the mean predictor. This is the failure mode that a random-split benchmark cannot detect, and it is exactly the failure mode that an experimenter designing a new stimulation paradigm would encounter.

The sequence models, by contrast, degrade gracefully on the same split (n/a for OPTOFORMER), because they read the stimulus directly rather than through summary statistics. The ordering is stable across all three OOD regimes and the paired per-trial comparisons are significant (Section 6).

5.3 Predictive uncertainty detects the shift

Deep ensembles over 3 seeds plus split-conformal prediction give a distribution-free interval at nominal 90% coverage, calibrated on the validation split. Coverage on the ID test is close to nominal (n/a for OPTOFORMER), while coverage degrades on the OOD splits, and the mean ensemble standard deviation increases ($n/a \rightarrow n/a$ spikes per trial). The conformal interval is therefore not only honest in-distribution, it also *warns*: a practitioner running protocol screening would see wide intervals and high ensemble disagreement exactly when the model is being extrapolated, which is the signal to fall back on a simpler protocol.

5.4 Few-shot transfer to a held-out cell type

We pretrained a model on two cell types and asked how much labelled data a third type needs. Zero-shot performance after removing that type from training is n/a ; fine-tuning on 50 labelled cells of the held-out type gives n/a , versus n/a when the same 50 cells are used for training from scratch and n/a for the oracle that saw the type during pretraining. Fine-tuning the pretrained model on the same 50 cells as the oracle (n/a) isolates the value of pretraining on other types from the value of the extra data. The pattern—pretraining helps most exactly when data are scarce—is the MAML-style intuition Finn et al. [2017], here obtained without any meta-learning machinery.

5.5 The model designs stimulation

We used the surrogate (ensemble mean of OPTOFORMER) as an objective inside a differential-evolution search over a ten-segment frequency schedule plus a global intensity, targeting a prescribed evoked spike count 20 spikes per trial. Three search strategies were compared (learned schedule, surrogate-optimised constant frequency, random search) plus an oracle constant-frequency protocol scanned directly in the simulator. Every candidate was finally scored in the ground-truth simulator on held-out cells of the target type, so the comparison measures design quality, not surrogate agreement.

Mean absolute error with respect to the target is n/a for schedule-based design, n/a for the constant-frequency surrogate and n/a for the simulator-scanned oracle, with 100% of held-out cells within ± 3 spikes of target for the designed protocols (n/a). The two conclusions that matter practically are: (i) a model trained only for prediction is a usable design surrogate, and (ii) the shape of the stimulus matters—protocols that a surrogate-optimised constant-frequency search cannot express outperform constant-frequency protocols of matched light dose.

5.6 Interpretability recovers biophysics

Gradient saliency concentrates on stimulus onsets: mean $|\partial \text{logit}/\partial L|$ inside pulse windows exceeds off-pulse saliency by n/a -fold for OPTOFORMER. Crucially, the negative control works: shuffling the light trace across trials destroys prediction (AUPRC falls to n/a , i.e. below the base rate of 0.008), demonstrating that the models use the stimulus rather than trial identity or cell identity alone.

The learned GLM filters recover cell-type-specific photocurrent kinetics: a fast onset and a slow decay, with differences between cell types consistent with their differing opsin expression and adaptation. Because the GLM kernel is a literal causal filter, its shape can be compared directly against the photocurrent time course used by the forward model, providing an independent check that the model has identified the right mechanism rather than a correlated shortcut.

6 Statistical analysis

All comparisons use the same held-out cells. Confidence intervals for R^2 , MAE and Spearman correlation are bootstrap percentile intervals over trials (1,000 resamples). Model comparisons use paired Wilcoxon signed-rank tests on per-trial absolute error with Holm correction across comparisons. Seed variability is reported as the standard deviation across 3 training seeds. Absolute effect sizes are reported alongside p -values, in keeping with the observation that with thousands of trials, statistical significance can accompany a practically trivial difference—so we emphasise the R^2 and MAE differences themselves.

7 Limitations and threats to validity

1. **Simulation.** The forward model, however carefully built, is not cortex. A model trained here has learned the simulator’s stimulus–response map. The transferable claim we make is about the *methodology* and about the failure modes, not about a specific prediction for a real neuron. Real-data validation should re-fit the light-to-spike map on patch-clamp or two-photon recordings and then apply the same OOD protocol to test whether the degradation pattern reported here reproduces.
2. **Single-neuron, single-photon class.** We model direct opsin-driven excitation of three cell types. Recurrent network interactions, opsin inhibition, and multi-wavelength designs are out of scope, though the simulator can be extended to them and the code makes this straightforward.
3. **Model scale.** The models here are small ($\sim 10^4$ – 10^5 parameters) because the benchmark is generated, not because small is optimal. We expect performance to improve with scale and with more protocol diversity; we report the data/parameter trade-off explicitly rather than tuning per model to a fixed compute budget that would bias the comparison.
4. **Frame-level stimulus lookahead.** CNN and LSTM operate on 8 ms frames and OPTOFORMER on 32 ms patches, so predictions may use stimulus information from within the current frame. Since protocols are specified in advance, this does not leak neuronal information; it does mean the models are not usable for reacting to a stimulus that is not known in advance, which matters only for truly online closed-loop use.

8 Conclusion

Optogenetics is limited less by our ability to deliver light than by our ability to predict what the light will do. We built a benchmark on which that question has a checkable answer, and showed

that (i) deep sequence models outperform both a classical encoding model and gradient-boosted trees on held-out cells, (ii) the classical tree approach fails precisely where novel stimulation paradigms live while sequence models do not, (iii) ensembles and conformal intervals expose that failure rather than hiding it, (iv) a few dozen cells adapt a model to a new cell type, and (v) a model trained only for prediction is a usable surrogate for designing stimulation. The practical consequence is a shorter experimental loop: screen, design, validate, repeat.

Data and code availability

All code, the data generator, configuration files, tests and result tables are released with this manuscript (repository: github.com/.../optonet). The dataset is regenerated deterministically from a single seed with `python scripts/01_generate_data.py`.

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